



1. รายละเอียดของโจทย์

1.1 ชื่อโจทย์ (ต้องเป็นภาษาอังกฤษ)

Cyber_hero

1.2 หมวดของโจทย์ (Red Team, Blue Team, Miscellaneous)

Blue Team

1.3 ชื่อทีม

0xBl4ck-c4t

1.4 ระดับการแข่งขัน (มัธยมศึกษา, อุดมศึกษา)

อุดมศึกษา

1.5 รายชื่อสมาชิกในทีม

หัวหน้าทีม: -

สมาชิกคนที่ 1: -

สมาชิกคนที่ 2: -

2. รายละเอียด Write-up และการแสดงขั้นตอนการแก้ปัญหาจากโจทย์

So first I inspected the file to see what could possibly be hidden inside the picture file.

I checked the file first to see what's the file format or anything hidden in.

command/tool: file

```
(base) (nothing@kali)~[~/Downloads]
$ file cyber_hero.jpeg
cyber_hero.jpeg: JPEG image data, JFIF standard 1.01, aspect ratio, density 1x1, segment length 16, baseline, precision 8, 447x447, component s 3
```

I used exiftool to inspect what metadata it contains.

command/tool: exiftool

```
(base) (nothing@kali)~[~/Downloads]
$ exiftool cyber_hero.jpeg
ExifTool Version Number      : 13.55
File Name                    : cyber_hero.jpeg
Directory                    : .
File Size                    : 29 kB
File Modification Date/Time   : 2026:09:11 22:37:09-04:00
File Access Date/Time        : 2026:09:11 22:37:20-04:00
File Inode Change Date/Time   : 2026:09:11 22:37:09-04:00
File Permissions              : -rw-rw-r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                  : 1.01
Resolution Unit               : None
X Resolution                  : 1
Y Resolution                  : 1
Image Width                   : 447
Image Height                  : 447
Encoding Process              : Baseline DCT, Huffman coding
Bits Per Sample               : 8
Color Components              : 3
Y Cb Cr Sub Sampling          : YCbCr4:2:0 (2 2)
Image Size                    : 447x447
Megapixels                    : 0.200
```

Nothing out of the ordinary.

I tried out the binwalk to see what's hiding the picture file or not.

Command/tool: binwalk

```
(base) (nothing@kali)-[~/Downloads]
└─$ binwalk cyber_hero.jpeg
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	JPEG image data, JFIF standard 1.01

So in the end I used the steghide tool to finally see what was really hidden in there.

Command / Tool: steghide

I checked out the info first and saw that there's embedded data which is the flag.txt.

```
(base) (nothing@kali)-[~/Downloads]
└─$ steghide info cyber_hero.jpeg
"cyber_hero.jpeg":
  format: jpeg
  capacity: 1.4 KB
Try to get information about embedded data ? (y/n) y
Enter passphrase:
  embedded file "flag.txt":
    size: 45.0 Byte
    encrypted: rijndael-128, cbc
    compressed: yes
```

I extracted the file by passing the passphrase empty.
steghide extract -sf cyber_hero.jpeg -p ""

And then read the flag to get it.

Command / too: steghide

```

(base) ~ (nothing@kali) [~/Downloads]
└─$ steghide extract -sf cyber_hero.jpeg -p ""
wrote extracted data to "flag.txt".

(base) ~ (nothing@kali) [~/Downloads]
└─$ ls
admin.json          cloudflared-linux-amd64.deb  flag.txt             live-art             payload.b64          rainier.jpg          volatility3
AWholeNewWorld.wld cookies.txt                handout.zip          my_os.png           payload.bin          srev                 ysoserial.jar
challenge.py        cyber_hero.jpeg            huge-binary-1        output.txt           p-equals-np.pdf    test2.json          test.json
chal.py            find_me                    larp                 out.txt             permissions

(base) ~ (nothing@kali) [~/Downloads]
└─$ cat flag.txt
CYBERHEROCTF{423945235349405354242495485040}

(base) ~ (nothing@kali) [~/Downloads]
└─$
└─$ steghide extract -sf cyber_hero.jpeg -p ""

```

Flag:

CYBERHEROCTF{423945235349405354242495485040}